

PHYS505 Recitation – Feb 11, 2025

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Problem 2.7

Potential problem in half space $z \geq 0$ with Dirichlet bc on the plane $z = 0$ (and at infinity)

(a) Write the Green function $G(\vec{x}, \vec{x}')$

$\Rightarrow$ We have this in the lecture notes – using the method of images (point charge next to a grounded infinite conducting plane), the Green function is ($4\pi\epsilon_0$ times) the potential at point $\vec{x}$ due to a unit point charge at point $\vec{x}'$ in the presence of the grounded plane

$$G(\vec{x}, \vec{x}') = \frac{1}{|\vec{x} - \vec{x}'|} - \frac{1}{|\vec{x} - \vec{x}''|} = \frac{1}{|\vec{x} - \vec{x}'|} - \frac{1}{|\vec{x} - \vec{x}' + 2z'\hat{z}|}$$

which includes the contribution from the image charge at $\vec{x}'' = \vec{x}' - 2z'\hat{z}$ with $z, z' > 0$.

(b) With the potential on the plane $z = 0$ specified as $\Phi = V$ inside a circle of radius a centered at the origin and $\Phi = 0$ outside, find an integral expression for $\Phi(\rho, \phi, z)$

$\Rightarrow$ Again, from the lecture notes (Green's theorem), the general solution (with the charge density $\rho(\vec{x}') = 0$ in the region of interest) is

$$\Phi(\vec{x}) = -\frac{1}{4\pi} \int_S \Phi(\vec{x}') \partial_{n'} G(\vec{x}, \vec{x}') da', \quad z > 0$$

With $\Phi(\vec{x}')$ given on the surface S (defined by $z' = 0$), we also need the normal derivative of G there. Cylindrical coordinates are appropriate for this problem due to the azimuthal symmetry

$$\begin{aligned} \vec{x} &= \rho \cos \phi \hat{x} + \rho \sin \phi \hat{y} + z \hat{z} \\ \vec{x}' &= \rho' \cos \phi' \hat{x} + \rho' \sin \phi' \hat{y} + z' \hat{z} \\ \vec{x}'' &= \rho' \cos \phi' \hat{x} + \rho' \sin \phi' \hat{y} - z' \hat{z} \end{aligned}$$

Writing the terms in the Green function in detail,

$$\begin{aligned} |\vec{x} - \vec{x}'|^2 &= (\vec{x} - \vec{x}') \cdot (\vec{x} - \vec{x}') = |\vec{x}|^2 + |\vec{x}'|^2 - 2\vec{x} \cdot \vec{x}' \\ &= \rho^2 + z^2 + \rho'^2 + z'^2 - 2(\rho \cos \phi \rho' \cos \phi' + \rho \sin \phi \rho' \sin \phi' + zz') \\ |\vec{x} - \vec{x}'|^{-1} &= [\rho^2 + z^2 + \rho'^2 + z'^2 - 2\rho\rho' \cos(\phi - \phi') - 2zz']^{-1/2} \\ |\vec{x} - \vec{x}''|^{-1} &= [\rho^2 + z^2 + \rho'^2 + z'^2 - 2\rho\rho' \cos(\phi - \phi') + 2zz']^{-1/2} \end{aligned}$$

and noting that the surface normal $\hat{n}'$ points out from the region where we are computing the solution

$$\hat{n}' = -\hat{z}'$$

we can evaluate the normal derivative

$$\begin{aligned}\partial_{z'}|\vec{x} - \vec{x}'|^{-1} &= (-1/2)|\vec{x} - \vec{x}'|^{-3}(2z' - 2z) = -\frac{z' - z}{|\vec{x} - \vec{x}'|^3} \\ \vec{\nabla}' \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) &= -\frac{\vec{x}' - \vec{x}}{|\vec{x} - \vec{x}'|^3} \\ \partial_{z'}|\vec{x} - \vec{x}''|^{-1} &= (-1/2)|\vec{x} - \vec{x}''|^{-3}(2z' + 2z) = -\frac{z' + z}{|\vec{x} - \vec{x}''|^3} \\ \partial_{n'}G(z' = 0) &= -\partial_{z'}G(z' = 0) = -\frac{2z}{[\rho^2 + z^2 + \rho'^2 - 2\rho\rho' \cos(\phi - \phi')]^{3/2}}\end{aligned}$$

which, when plugged into the integral for the potential, gives

$$\Phi(\rho, \phi, z > 0) = \frac{Vz}{2\pi} \int_0^{2\pi} d\phi' \int_0^a d\rho' \frac{\rho'}{[\rho^2 + z^2 + \rho'^2 - 2\rho\rho' \cos(\phi - \phi')]^{3/2}}$$

While this is a valid expression for the solution, it doesn't clearly reflect the symmetry of the problem which demands

$$\partial_\phi \Phi = 0, \quad \Phi = \Phi(\rho, z)$$

Therefore, we can simply assign a particular value, eg. $\phi = 0$, in the formula (equivalent to a change of variable for the angular integration to $\phi' - \phi$)

$$\boxed{\Phi(\rho, z > 0) = \frac{Vz}{2\pi} \int_0^{2\pi} d\phi' \int_0^a d\rho' \frac{\rho'}{[\rho'^2 - 2\rho\rho' \cos \phi' + z^2 + \rho^2]^{3/2}}}$$

Note that the ϕ' integral runs through a full period of $\cos(\phi - \phi')$ independent of the value of ϕ .

(c) Show that the potential along the z -axis is

$$\Phi = V \left(1 - \frac{z}{\sqrt{a^2 + z^2}} \right)$$

$\Rightarrow$ Plugging $\rho = 0$ into the potential formula in (b)

$$\begin{aligned}\Phi(\rho = 0, z > 0) &= \frac{Vz}{2\pi} \int_0^{2\pi} d\phi' \int_0^a d\rho' \frac{\rho'}{[z^2 + \rho'^2]^{3/2}} = Vz \int_0^a d\rho' \frac{\rho'}{[z^2 + \rho'^2]^{3/2}} \\ u &= \rho'^2 + z^2, \quad du = 2\rho' d\rho' \\ \Phi(0, z > 0) &= \frac{Vz}{2} \int u^{-3/2} du = Vz (\rho'^2 + z^2)^{-1/2} \Big|_0^a = Vz \left(\frac{1}{z} - \frac{1}{\sqrt{a^2 + z^2}} \right)\end{aligned}$$

which yields

$$\boxed{\Phi(0, z > 0) = V \left(1 - \frac{z}{\sqrt{a^2 + z^2}} \right)}$$

(d) Show that at large distances ($\rho^2 + z^2 \gg a^2$), the potential can be expanded in a power series in $(\rho^2 + z^2)^{-1}$ with leading terms

$$\Phi = \frac{Va^2}{2} \frac{z}{(\rho^2 + z^2)^{3/2}} \left[1 - \frac{3a^2}{4(\rho^2 + z^2)} + \frac{5(3\rho^2 a^2 + a^4)}{8(\rho^2 + z^2)^2} + \dots \right]$$

and verify that it's consistent with the result in (c).

$\Rightarrow$ Starting from the expression for the potential in part (b)

$$\begin{aligned}\Phi(\rho, z) &= \frac{Vz}{2\pi} \int_0^{2\pi} d\phi \int_0^a d\rho' \frac{\rho'}{[r^2 + \rho'^2 - 2\rho\rho' \cos \phi]^{3/2}} \\ r^2 &= \rho^2 + z^2\end{aligned}$$

and using variable substitution

$$\begin{aligned}
\rho' &= a u \\
[r^2 + \rho'^2 - 2\rho\rho' \cos \phi]^{3/2} &= r^3 \left[1 + \frac{a^2}{r^2} u^2 - \frac{2a\rho}{r^2} u \cos \phi \right]^{3/2} \\
\Phi(\rho, z) &= \frac{V a^2}{2\pi} \frac{z}{r^3} \int_0^{2\pi} d\phi \int_0^1 \frac{u du}{\left[1 + \frac{a^2}{r^2} u^2 - \frac{2a\rho}{r^2} u \cos \phi \right]^{3/2}} \\
&= \frac{V a^2}{2\pi} \frac{z}{r^3} I(\rho, z)
\end{aligned}$$

We are interested in the case where

$$\alpha = \frac{a^2}{\rho^2 + z^2} = \left(\frac{a}{r} \right)^2 \ll 1$$

Note that u , I and α are dimensionless. We expand the integrand in powers of α using a Mclaurin series

$$\begin{aligned}
f(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots \\
(1+x)^\beta &= 1 + \beta x + \frac{\beta(\beta-1)}{2}x^2 + \frac{\beta(\beta-1)(\beta-2)}{6}x^3 + \dots \\
(1+x)^{-3/2} &= 1 - \frac{3}{2}x + \frac{15}{8}x^2 + \dots \\
\left[1 + \frac{a^2}{r^2}u^2 - \frac{2a\rho}{r^2}u \cos \phi \right]^{-3/2} &= \left[1 + \alpha \left(u^2 - 2\frac{\rho}{a} \cos \phi u \right) \right]^{-3/2} \\
&= 1 - \frac{3}{2}\alpha \left(u^2 - 2\frac{\rho}{a} \cos \phi u \right) + \frac{15}{8} \left[\alpha \left(u^2 - 2\frac{\rho}{a} \cos \phi u \right) \right]^2 + \mathcal{O}(\alpha^3)
\end{aligned}$$

and keep terms up to 2nd order in α to evaluate the integral I

$$\begin{aligned}
I &= \int_0^{2\pi} d\phi \int_0^1 du u \left\{ 1 - \frac{3}{2}\alpha \left(u^2 - 2\frac{\rho}{a} \cos \phi u \right) + \frac{15}{8} \left[\alpha \left(u^2 - 2\frac{\rho}{a} \cos \phi u \right) \right]^2 \right\} + \mathcal{O}(\alpha^3) \\
&= \int_0^{2\pi} d\phi \int_0^1 du \left\{ u - \frac{3}{2}\alpha \left(u^3 - 2\frac{\rho}{a} \cos \phi u^2 \right) \right. \\
&\quad \left. + \frac{15}{8}\alpha^2 \left[u^5 - 4\frac{\rho}{a} \cos \phi u^4 + 4 \left(\frac{\rho}{a} \right)^2 \cos^2 \phi u^3 \right] \right\} + \mathcal{O}(\alpha^3) \\
&= \int_0^{2\pi} d\phi \left[\frac{1}{2} - \frac{3}{2}\alpha \left(\frac{1}{4} - 2\frac{\rho}{a} \cos \phi \frac{1}{3} \right) \right] \\
&\quad + \frac{15}{8}\alpha^2 \int_0^{2\pi} d\phi \left[\frac{1}{6} - 4\frac{\rho}{a} \cos \phi \frac{1}{5} + 4 \left(\frac{\rho}{a} \right)^2 \cos^2 \phi \frac{1}{4} \right] + \mathcal{O}(\alpha^3) \\
&\simeq \int_0^{2\pi} d\phi \left[\left(\frac{1}{2} - \frac{3}{8}\alpha + \frac{5}{16}\alpha^2 \right) - \left(1 + \frac{3}{2}\alpha^2 \right) \frac{\rho}{a} \cos \phi + \frac{15}{8}\alpha^2 \left(\frac{\rho}{a} \right)^2 \cos^2 \phi \right] \\
&= \left(\frac{1}{2} - \frac{3}{8}\alpha + \frac{5}{16}\alpha^2 \right) 2\pi + \frac{15}{8}\alpha^2 \left(\frac{\rho}{a} \right)^2 \pi + \mathcal{O}(\alpha^3) \\
&= 2\pi \left\{ \frac{1}{2} - \frac{3}{8}\alpha + \frac{5}{16} \left[1 + 3 \left(\frac{\rho}{a} \right)^2 \right] \alpha^2 \right\} + \mathcal{O}(\alpha^3)
\end{aligned}$$

which, when substituted into the expression for the potential, gives

$$\begin{aligned}
\Phi &= \frac{V a^2}{2\pi} \frac{z}{r^3} I = V a^2 \frac{z}{r^3} \left\{ \frac{1}{2} - \frac{3}{8}\alpha + \frac{5}{16} \left[1 + 3 \left(\frac{\rho}{a} \right)^2 \right] \alpha^2 \right\} + \mathcal{O}(\alpha^3) \\
&= \frac{V a^2}{2} \frac{z}{r^3} \left\{ 1 - \frac{3}{4}\alpha + \frac{5}{8} \left[1 + 3 \left(\frac{\rho}{a} \right)^2 \right] \alpha^2 \right\} + \mathcal{O}(\alpha^3)
\end{aligned}$$

or, after substituting for α

$$\Phi(\rho, z) = \frac{Va^2}{2} \frac{z}{(\rho^2 + z^2)^{3/2}} \left\{ 1 - \frac{3a^2}{4(\rho^2 + z^2)} + \frac{5(3\rho^2 a^2 + a^4)}{8(\rho^2 + z^2)^2} + \dots \right\}$$

Next, we consider what happens to this expression on-axis ($\rho = 0$)

$$\begin{aligned} \Phi_d(0, z \gg a) &= \frac{Va^2}{2} \frac{z}{(z^2)^{3/2}} \left[1 - \frac{3}{4} \frac{a^2}{z^2} + \frac{5}{8} \frac{a^4}{(z^2)^2} + \dots \right] \\ &= \frac{V}{2} \left(\frac{a}{z} \right)^2 \left[1 - \frac{3}{4} \left(\frac{a}{z} \right)^2 + \frac{5}{8} \left(\frac{a}{z} \right)^4 + \dots \right] \end{aligned}$$

and compare to the result in part (c)

$$\begin{aligned} \Phi_c(0, z) &= V \left(1 - \frac{z}{\sqrt{a^2 + z^2}} \right) = V \left\{ 1 - \left[1 + \left(\frac{a}{z} \right)^2 \right]^{-1/2} \right\} \\ \left[1 + \left(\frac{a}{z} \right)^2 \right]^{-1/2} &= 1 - \frac{1}{2} \left(\frac{a}{z} \right)^2 + \frac{3}{8} \left(\frac{a}{z} \right)^4 - \frac{5}{16} \left(\frac{a}{z} \right)^6 + \mathcal{O} \left(\left(\frac{a}{z} \right)^8 \right) \\ \Phi_c(0, z \gg a) &= V \left[\frac{1}{2} \left(\frac{a}{z} \right)^2 - \frac{3}{8} \left(\frac{a}{z} \right)^4 + \frac{5}{16} \left(\frac{a}{z} \right)^6 + \dots \right] \\ &= \frac{V}{2} \left(\frac{a}{z} \right)^2 \left[1 - \frac{3}{4} \left(\frac{a}{z} \right)^2 + \frac{5}{8} \left(\frac{a}{z} \right)^4 + \dots \right] \end{aligned}$$

Problem 3.2

Spherical surface of radius R with uniform $\sigma = Q/(4\pi R^2)$ and missing cap of cone angle α at the north pole.

(a) show that the potential inside is

$$\Phi = \frac{Q}{8\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} [P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)] \frac{r^l}{R^{l+1}} P_l(\cos \theta)$$

with the definition $P_{-1}(\cos \alpha) = -1$. Find the potential outside.

(b) Find $\vec{E}(\vec{0})$.

(c) Discuss $\lim_{\alpha \rightarrow 0, \pi} \Phi, \vec{E}$

(a) Azimuthal symmetry means the potential can be written as

$$\begin{aligned} \Phi(r, \theta) &= \sum_{l=0}^{\infty} \left[A_l r^l + B_l r^{-(l+1)} \right] P_l(\cos \theta) \\ &= A_0 + A_1 r P_1(\cos \theta) + A_2 r^2 P_2(\cos \theta) + \dots \\ &\quad + \frac{B_0}{r} + \frac{B_1}{r^2} P_1(\cos \theta) + \frac{B_2}{r^3} P_2(\cos \theta) + \dots \end{aligned}$$

In order for $\Phi_{out}(r \rightarrow \infty)$ and $\Phi_{in}(0)$ not to blow up, we need to ditch the terms with +ve powers of r outside and -ve powers of r inside

$$\begin{aligned} \Phi_{out}(r > R, \theta) &= A'_0 + \sum_{l=0}^{\infty} B_l r^{-(l+1)} P_l(\cos \theta) \\ \Phi_{in}(r < R, \theta) &= \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta) \end{aligned}$$

and the solutions should match on the spherical boundary

$$\Phi_{out}(R) = \Phi_{in}(R)$$

For this compact charge distribution, the potential at $r \rightarrow \infty$ should look like that from a point charge

$$\begin{aligned} \Phi_{out}(r \gg R) &\rightarrow A'_0 + \frac{B_0}{r} = \frac{q}{4\pi\epsilon_0 r} \\ \Rightarrow A'_0 &= 0, \quad B_0 = \frac{q}{4\pi\epsilon_0} \end{aligned}$$

where q is the total charge on the surface spanned by $\theta \in [\alpha, \pi]$

$$\begin{aligned} q &= \int \sigma da = R^2 \int_0^{2\pi} d\phi \int_{\alpha}^{\pi} d\theta \sin \theta \frac{Q}{4\pi R^2} \\ &= R^2 (2\pi) \frac{Q}{4\pi R^2} \cos \theta|_{\pi}^{\alpha} = \frac{Q}{2} (1 + \cos \alpha) \end{aligned}$$

Going back to the potential on the surface for the rest of the coefficients

$$\begin{aligned} \Phi_{in}(r = R, \theta) &= A_0 + \sum_{l=1}^{\infty} A_l R^l P_l(\cos \theta) \\ \Phi_{out}(r = R, \theta) &= \frac{q}{4\pi\epsilon_0 R} + \sum_{l=1}^{\infty} \frac{B_l}{R^{l+1}} P_l(\cos \theta) \end{aligned}$$

Since the Legendre expansion of the potential function on the surface is unique, every pair of coefficients must be the same

$$A_0 = \frac{q}{4\pi\epsilon_0 R} = \frac{Q}{8\pi\epsilon_0 R}(1 + \cos \alpha)$$

$$B_l = R^{2l+1}A_l, \quad l > 0$$

which gives us a relation between the two sets of coefficients but is not enough to determine all of them. The other piece of information we need to use is the charge density σ which sets the discontinuity in the electric field on the surface

$$\begin{aligned} \frac{\sigma}{\epsilon_0} &= \hat{r} \cdot (\vec{E}_{out} - \vec{E}_{in})_{r=R} \\ &= (\partial_r \Phi_{in} - \partial_r \Phi_{out})_{r=R} \\ &= \frac{q}{4\pi\epsilon_0 R^2} + \sum_{l=1}^{\infty} \left[lA_l R^{l-1} + \frac{(l+1)B_l}{R^{l+2}} \right] P_l(\cos \theta) \end{aligned}$$

Plugging in the previous relation between the A_l 's and B_l 's,

$$\sigma(\theta) = \frac{q}{4\pi R^2} + \epsilon_0 \sum_{l=1}^{\infty} (2l+1)A_l R^{l-1} P_l(\cos \theta)$$

Again, due to the uniqueness of the Legendre series, the expansion coefficients must match. We need to build the Legendre expansion for the charge density in order to solve for the coefficients

$$\sigma(\theta) = \sum_{l=0}^{\infty} S_l P_l(\cos \theta)$$

We employ the orthogonality relation

$$\int_{-1}^1 P_l(\cos \theta) P_{l'}(\cos \theta) d(\cos \theta) = \frac{2}{2l+1} \delta_{ll'}$$

to find the coefficients S_l for $l > 0$

$$\begin{aligned} \int_{-1}^1 d(\cos \theta) P_l(\cos \theta) \sigma(\theta) &= \frac{2}{2l+1} S_l \\ \frac{Q}{4\pi R^2} \int_{-1}^{\cos \alpha} d(\cos \theta) P_l(\cos \theta) &= \frac{2}{2l+1} S_l \end{aligned}$$

The integral can be evaluated by using the derivative recursion relation for Legendre polynomials

$$\begin{aligned} \frac{d}{dx} [P_{l+1}(x) - P_{l-1}(x)] &= (2l+1)P_l(x) \\ [P_{l+1}(x) - P_{l-1}(x)]_{-1}^{\cos \alpha} &= (2l+1) \int_{-1}^{\cos \alpha} P_l(x) dx \end{aligned}$$

We can now match the coefficients for $l \geq 1$ to get

$$\begin{aligned} \epsilon_0(2l+1)A_l R^{l-1} &= \frac{Q}{8\pi R^2} [P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)] \\ A_l &= \frac{Q}{8\pi\epsilon_0} \frac{1}{R^{l+1}} \frac{1}{2l+1} [P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)] \\ B_l &= \frac{Q}{8\pi\epsilon_0} \frac{R^l}{2l+1} [P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)] \end{aligned}$$

which yields

$$S_l = \frac{Q}{8\pi R^2} [P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)], \quad l > 0$$

and plug them into the expression for the potential inside

$$\begin{aligned} \Phi_{in}(r, \theta) &= A_0 + \sum_{l=1}^{\infty} A_l r^l P_l(\cos \theta) \\ &= \frac{Q}{8\pi\epsilon_0 R} \left[(1 + \cos \alpha) + \sum_{l=1}^{\infty} \frac{P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)}{2l + 1} \left(\frac{r}{R}\right)^l P_l(\cos \theta) \right] \end{aligned}$$

It seems unnecessary to try and make this slightly more palatable with some arbitrary definition but we'll indulge Prof. Jackson and observe that

$$\begin{aligned} P_1(\cos \alpha) &= \cos \alpha, \quad P_{-1}(\cos \alpha) \equiv -1 \\ \Rightarrow 1 + \cos \alpha &= \frac{P_1(\cos \alpha) - P_{-1}(\cos \alpha)}{2 \cdot 0 + 1} \left(\frac{r}{R}\right)^0 P_0(\cos \theta) \\ \Phi_{in}(r, \theta) &= \frac{Q}{8\pi\epsilon_0 R} \sum_{l=0}^{\infty} \frac{P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)}{2l + 1} \left(\frac{r}{R}\right)^l P_l(\cos \theta) \end{aligned}$$

which leads to the simpler looking expression

$$\boxed{\Phi_{in}(r, \theta) = \frac{Q}{8\pi\epsilon_0 R} \sum_{l=0}^{\infty} \frac{P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)}{2l + 1} \left(\frac{r}{R}\right)^l P_l(\cos \theta)}$$

The potential outside is

$$\begin{aligned} \Phi_{out}(r, \theta) &= \frac{q}{4\pi\epsilon_0 r} + \sum_{l=1}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta) \\ &= \frac{Q}{8\pi\epsilon_0 r} \left[1 + \cos \alpha + \sum_{l=1}^{\infty} \frac{P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)}{2l + 1} \left(\frac{R}{r}\right)^l P_l(\cos \theta) \right] \end{aligned}$$

or, with the definition above,

$$\boxed{\Phi_{out}(r, \theta) = -\frac{Q}{8\pi\epsilon_0 r} \sum_{l=0}^{\infty} \frac{P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)}{2l + 1} \left(\frac{R}{r}\right)^l P_l(\cos \theta)}$$

(b) We can do this the easy way or the hard way (always wanted to work that into some class material :-). The easy (physicsy) way is by using superposition since the problem is equivalent to a closed sphere with constant charge density $\sigma_1 = \sigma$ (which gives 0 field at the origin) plus a spherical cap of cone angle α with charge density $\sigma_2 = -\sigma$. If we take a portion of the cap with area da and charge dq , its contribution to the field $\vec{E}_0$ at the origin is given by

$$\begin{aligned} d\vec{E}_0 &= \frac{dq}{4\pi\epsilon_0 r^2} (-\hat{r}) = \frac{\sigma_2 da}{4\pi\epsilon_0 R^2} (-\hat{r}) \\ &= \frac{\sigma}{4\pi\epsilon_0 R^2} da \hat{r} \end{aligned}$$

Due to azimuthal symmetry, only the z component of the field will survive

$$\begin{aligned} \hat{z} \cdot d\vec{E}_0 &= \frac{\sigma}{4\pi\epsilon_0 R^2} da (\hat{z} \cdot \hat{r}) \\ dE_z &= \frac{\sigma}{4\pi\epsilon_0 R^2} (2\pi R^2 \sin \theta d\theta) (\cos \theta) \end{aligned}$$

and we can integrate over the cap to find

$$\begin{aligned} E_z &= \int dE_z = \frac{\sigma}{2\epsilon_0} \int_{-1}^{\cos \alpha} \cos \theta d(-\cos \theta) \\ &= \frac{\sigma}{2\epsilon_0} \left[\frac{(\cos \theta)^2}{2} \right]_{\cos \alpha}^{-1} = \frac{\sigma}{4\epsilon_0} (1 - \cos^2 \alpha) \end{aligned}$$

Substituting for σ , we have

$$\boxed{\vec{E}_0 = \frac{Q \sin^2 \alpha}{16\pi\epsilon_0 R^2} \hat{z}}$$

The hard (mathy) way is by taking the gradient of the potential,

$$\begin{aligned} \vec{E}_0 &= -\vec{\nabla} \Phi_{in} \Big|_{r=0} \\ &= - \left(\hat{r} \partial_r + \hat{\theta} \frac{1}{r} \partial_\theta + \hat{\phi} \frac{1}{r \sin \theta} \partial_\phi \right) \Phi_{in}(r, \theta) \Big|_{r=0} \\ \Phi_{in}(r, \theta) &= A_0 + A_1 r P_1(\cos \theta) + A_2 r^2 P_2(\cos \theta) + \dots \end{aligned}$$

There is no azimuthal component due to symmetry (the ϕ -derivative is 0), the constant term ($l = 0$) in the potential does not contribute (all the derivatives are 0) and the substitution $r = 0$ gets rid of any term with a power of r left in it after the derivative, so only the $l = 1$ term remains

$$\begin{aligned} \vec{E}_0 &= -A_1 P_1(\cos \theta) \hat{r} - A_1 P_1'(\cos \theta) \hat{\theta} \Big|_{r=0} \\ &= -A_1 \left(\cos \theta \hat{r} - \sin \theta \hat{\theta} \right) \Big|_{r=0} \end{aligned}$$

This is a very simple but useless result because except for the constant A_1 , none of the factors make sense at $r = 0$ – we don't know what to substitute for θ , $\hat{r}$, or $\hat{\theta}$! If we approach the origin along the symmetry axis, there is no confusion

$$\vec{E}_0 = \lim_{z \rightarrow 0} \vec{E}(r = z, \theta = 0) = -A_1 \hat{z}$$

and of course we end up with the same answer,

$$\begin{aligned} A_1 &= \frac{Q}{8\pi\epsilon_0} \frac{1}{R^2} \frac{1}{3} [P_2(\cos \alpha) - P_0(\cos \alpha)] \\ &= \frac{Q}{24\pi\epsilon_0 R^2} \left[\frac{1}{2} (3 \cos^2 \alpha - 1) - 1 \right] \\ &= -\frac{Q \sin^2 \alpha}{16\pi\epsilon_0 R^2} \end{aligned}$$

(c) In the limit $\alpha \rightarrow 0$, the missing cap shrinks to a point at the north pole and we have a full spherical surface with uniform charge density, so we expect the field/potential of a point charge Q at the origin outside and no electric field and constant potential inside (Gauss' law)

$$\begin{aligned} \Phi_{out} &= \frac{Q}{4\pi\epsilon_0 r} \\ \Phi_{in} &= \frac{Q}{4\pi\epsilon_0 R} \\ \vec{E}_0 &= \vec{0} \end{aligned}$$

Looking at the solution as $\cos \alpha \rightarrow 1$, all the coefficients in the potential except $l = 0$ vanish since $P_l(1) = 1$, $q \rightarrow Q$, and we are left with

$$\begin{aligned}\Phi_{out} &\rightarrow \frac{B_0}{r} = \frac{Q}{4\pi\epsilon_0 r} \\ \Phi_{in} &\rightarrow A_0 = \frac{Q}{4\pi\epsilon_0 R} \\ \vec{E}_0 &= -A_1 \hat{z} \rightarrow \vec{0}\end{aligned}$$

The hard limit gives the correct answer. Next, we explore the explicit dependence on α which enters through the coefficients. Using a Taylor series expansion,

$$\begin{aligned}\cos \alpha &= 1 - \frac{\alpha^2}{2} + \mathcal{O}(\alpha^4) \\ q &= Q \left[1 - \frac{\alpha^2}{4} + \mathcal{O}(\alpha^4) \right] \\ A_0 &= \frac{q}{4\pi\epsilon_0 R} = \frac{Q}{4\pi\epsilon_0 R} \left(1 - \frac{\alpha^2}{4} \right) + \mathcal{O}(\alpha^4) \\ B_0 &= \frac{Q}{4\pi\epsilon_0} \left(1 - \frac{\alpha^2}{4} \right) + \mathcal{O}(\alpha^4)\end{aligned}$$

and we need to work a little harder for $l > 0$

$$\begin{aligned}P_l(\cos \alpha) &\simeq P_l(1 - \alpha^2/2), \quad l > 0 \\ &= P_l(1) - P'_l(1) \frac{\alpha^2}{2} + \mathcal{O}(\alpha^4) \\ P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha) &= - [P'_{l+1}(1) - P'_{l-1}(1)] \frac{\alpha^2}{2} + \mathcal{O}(\alpha^4)\end{aligned}$$

Using the derivative recursion formula again,

$$\begin{aligned}P'_{l+1}(1) - P'_{l-1}(1) &= (2l+1)P_l(1) \\ P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha) &= -\frac{2l+1}{2}\alpha^2 + \mathcal{O}(\alpha^4), \quad l > 0\end{aligned}$$

and we finally have the coefficients

$$\begin{aligned}A_l &= \frac{Q}{8\pi\epsilon_0} \frac{1}{R^{l+1}} \frac{1}{2l+1} [P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)] \\ &\simeq \frac{Q}{16\pi\epsilon_0 R^{l+1}} \alpha^2, \quad l > 0 \\ B_l &\simeq \frac{QR^l}{16\pi\epsilon_0} \alpha^2\end{aligned}$$

to plug into the potential formulas

$$\begin{aligned}\Phi_{out} &= \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta) \simeq \frac{Q}{16\pi\epsilon_0} \left[\frac{4 - \alpha^2}{r} + \alpha^2 \sum_{l=1}^{\infty} \frac{R^l}{r^{l+1}} P_l(\cos \theta) \right] \\ \Phi_{in} &= \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta) \simeq \frac{Q}{16\pi\epsilon_0} \left[4 - \frac{\alpha^2}{2} + \alpha^2 \sum_{l=1}^{\infty} \frac{r^l}{R^{l+1}} P_l(\cos \theta) \right] \\ \vec{E}_0 &= -A_1 \hat{z} \simeq -\frac{\alpha^2}{4} \frac{Q}{4\pi\epsilon_0 R} \hat{z}\end{aligned}$$

If we recall the Legendre expansion for the free-space Green function

$$\frac{1}{|\vec{r} - \vec{r}'|} = \frac{1}{r_{>}} \sum_{l=0}^{\infty} \left(\frac{r_{<}}{r_{>}} \right)^l P_l(\cos \gamma), \quad \cos \gamma = \frac{\vec{r} \cdot \vec{r}'}{r r'}$$

and rearrange the results slightly

$$\begin{aligned} \vec{r} &= r \hat{r}, \quad \vec{r}' = R \hat{z} \Rightarrow \gamma = \theta \\ \Phi_{out} &= \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{r} - \frac{\alpha^2}{4} \frac{1}{r} \sum_{l=0}^{\infty} \left(\frac{R}{r} \right)^l P_l(\cos \theta) \right] = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{r} - \frac{\alpha^2}{4} \frac{1}{|\vec{r} - R \hat{z}|} \right] \\ \Phi_{in} &= \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{R} - \frac{\alpha^2}{4} \frac{1}{R} \sum_{l=0}^{\infty} \left(\frac{r}{R} \right)^l P_l(\cos \theta) \right] = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{R} - \frac{\alpha^2}{4} \frac{1}{|\vec{r} - R \hat{z}|} \right] \end{aligned}$$

the physical meaning becomes clear – we have a superposition of two terms

- the potential equivalent to that from a point charge Q at the origin, representing the approximately complete sphere of uniform charge density
- a point charge $-(Q - q) = -\alpha^2 Q/4$ at the north pole, representing the missing cap

In the other limit $\alpha \rightarrow \pi$, the solution should look like that from a single point charge at the south pole

$$\begin{aligned} q &\simeq Q \left(\frac{\pi - \alpha}{2} \right)^2 \\ \Phi &\simeq \frac{q}{4\pi\epsilon_0} \frac{1}{|\vec{r} + R \hat{z}|} \\ \vec{E}_0 &\simeq \frac{q}{4\pi\epsilon_0 R^2} \hat{z} \end{aligned}$$

It's straightforward to follow the same steps by expanding $\cos \alpha$ in $\delta = \pi - \alpha$,

$$\begin{aligned} \cos \alpha &= -1 + \frac{\delta^2}{2} + \mathcal{O}(\delta^4), \quad q = Q \left[\frac{\delta^2}{4} + \mathcal{O}(\delta^4) \right] \\ A_0 &\simeq \frac{Q}{4\pi\epsilon_0 R} \frac{\delta^2}{4}, \quad B_0 \simeq \frac{Q}{4\pi\epsilon_0} \frac{\delta^2}{4} \end{aligned}$$

and $P_l(\cos \alpha)$ around -1

$$\begin{aligned} P_l(\cos \alpha) &\simeq P_l(-1 + \delta^2/2) \\ &= (-1)^l + P'_l(-1) \frac{\delta^2}{2}, \quad l > 0 \\ P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha) &\simeq [P'_{l+1}(-1) - P'_{l-1}(-1)] \frac{\delta^2}{2} \\ &\simeq \frac{2l+1}{2} (-1)^l \frac{\delta^2}{2} \end{aligned}$$

to get the coefficients

$$\begin{aligned} A_l &\simeq \frac{Q}{4\pi\epsilon_0} \frac{(-1)^l \delta^2}{R^{l+1} 4} \\ B_l &\simeq \frac{Q}{4\pi\epsilon_0} (-1)^l R^l \frac{\delta^2}{4} \end{aligned}$$

and the potential and E-field

$$\begin{aligned}
\vec{r} &= r \hat{r}, \quad \vec{r}' = -R \hat{z} \Rightarrow \gamma = \pi - \theta, \quad \cos \gamma = -\cos \theta \\
P_l(-\cos \theta) &= (-1)^l P_l(\cos \theta) \\
\Phi_{out} &\simeq \frac{\delta^2}{4} \frac{Q}{4\pi\epsilon_0} \frac{1}{r} \sum_{l=0}^{\infty} \left(\frac{r}{R}\right)^l P_l(-\cos \theta) = \frac{\delta^2}{4} \frac{Q}{4\pi\epsilon_0} \frac{1}{|\vec{r} + R \hat{z}|} \\
\Phi_{in} &\simeq \frac{\delta^2}{4} \frac{Q}{4\pi\epsilon_0} \frac{1}{R} \sum_{l=0}^{\infty} \left(\frac{R}{r}\right)^l P_l(-\cos \theta) = \frac{\delta^2}{4} \frac{Q}{4\pi\epsilon_0} \frac{1}{|\vec{r} + R \hat{z}|} \\
\vec{E}_0 &\simeq \frac{\delta^2}{4} \frac{Q}{4\pi\epsilon_0 R^2} \hat{z}
\end{aligned}$$